

Module 4: Case Study - 1

-By MG Ishaque

Problem Statement:

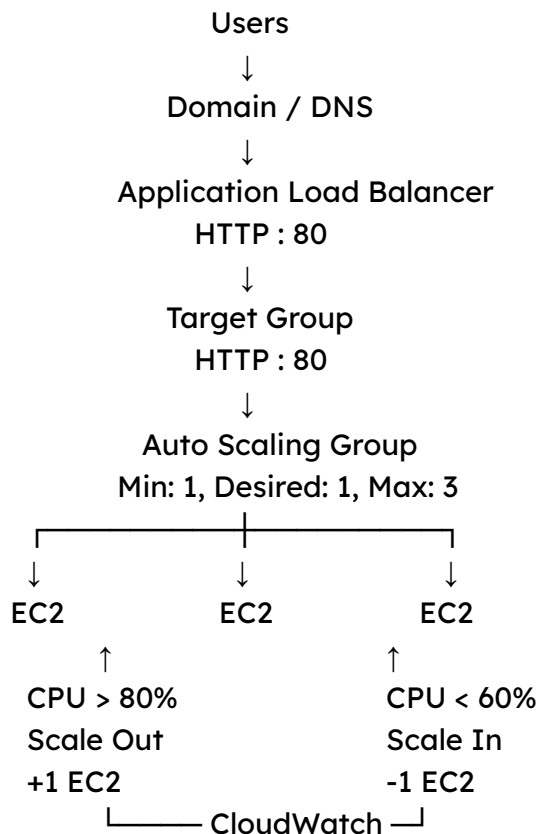
You work for XYZ Corporation that uses on premise solutions and a limited number of systems. With the increase in requests in their application, the load also increases. So, to handle the load the corporation has to buy more systems almost on a regular basis. Realizing the need to cut down the expenses on systems, they decided to move their infrastructure to AWS.

Tasks To Do:

1. To manage the scaling requirements of the company by:
 - a. Deploying multiple compute resources on the cloud as soon as the load increases and the CPU utilization exceeds 80%.
 - b. Removing the resources when the CPU utilization goes under 60%.
2. To create a load balancer to distribute the load between compute resources.
3. To route the traffic to the company's domain.

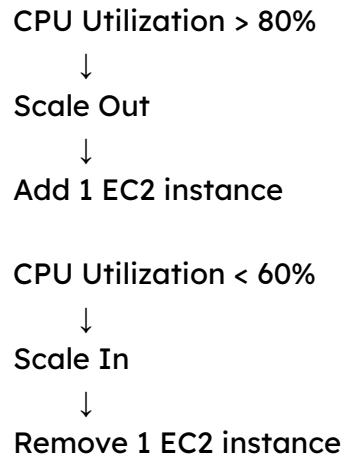
Architecture / Strategy:

Architecture:



Scaling logic: CPU > 80% → Scale OUT (+1 EC2)
CPU < 60% → Scale IN (-1 EC2)

Cloud watch strategy:



Task Index:

Task 1:

- Steps to create EC2 Web Server #1
- Steps to create AMI
- Steps to create Launch Template

Task 2:

- Steps to create Auto Scaling Group
- Steps to create Target Group
- Steps to create Application Load Balancer (ALB)
- Steps to connect ALB to Target Group

Task 3: Configured Dynamic auto Scale-In/Out using Cloudwatch alarm

- Steps to Scale out policy with CloudWatch CPU alarm
- Steps to Scale in policy with CloudWatch CPU alarm

Task 4: Routing to Domain

- Created a public Route 53 hosted zone.
- Added A record to Route 53 hosted zone

Task 5: Verification

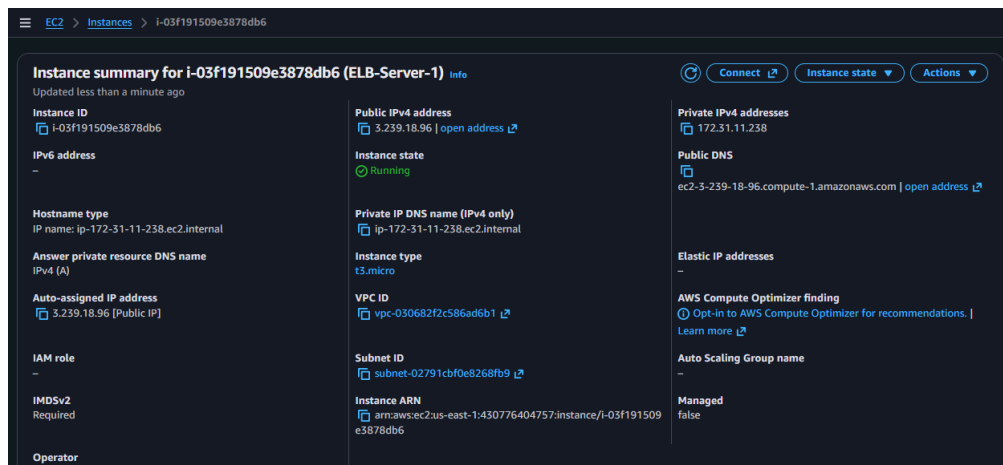
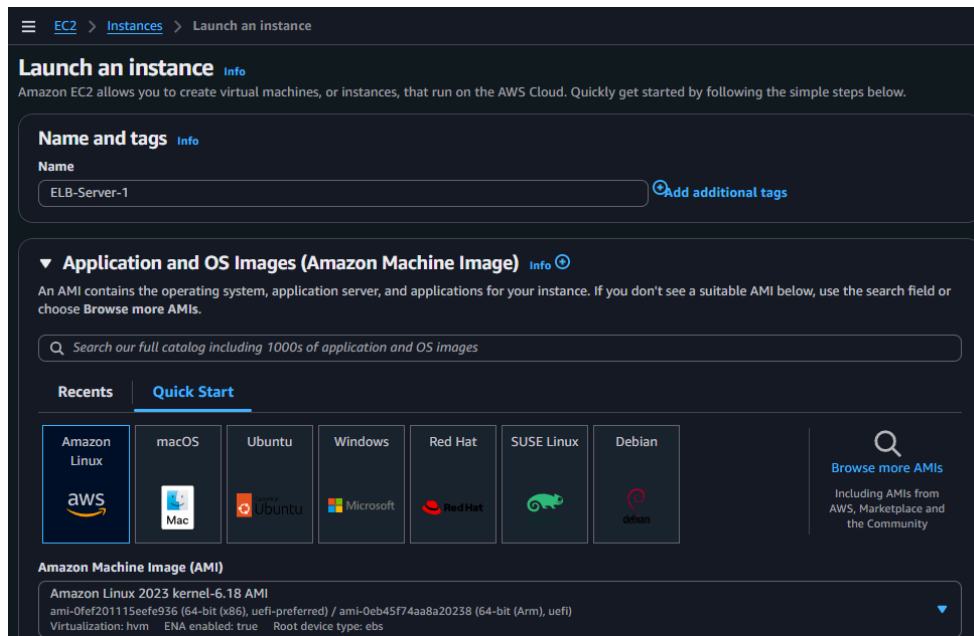
Task 1:

A. Created EC2 instance

EC2 → Instances → Launch instance

Selected Configuration and launched 1 EC2 server.

ELB-Server-1



Installed Web-Servers in the EC2 servers

Connected to ELB-Server-1

ELB-Server-1 → Connect

EC2 Instance Connect → Connect

Ran the following bash to install Apache webserver and enable it:

On terminal ran:

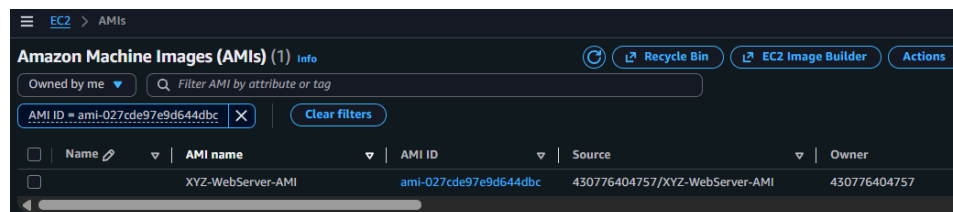
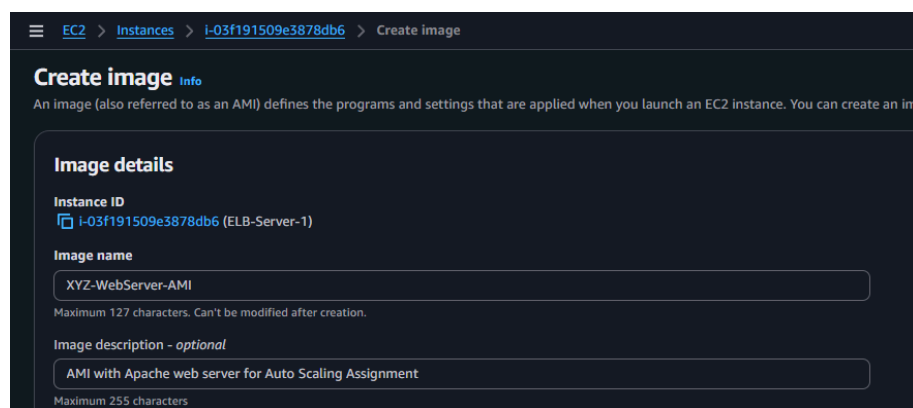
```
sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
```


C. Created the AMI

1. Selected **ELB-Server-1**
2. Clicked **Actions**
3. Selected: **Image and templates** → **Create image**

Configured it as follows:

- Image name: **XYZ-WebServer-AMI**
- Image description: “**AMI with Apache web server for Auto Scaling Assignment**”



D. Created a launch Template with this AMI.

Steps and configurations :

EC2 → Launch Templates → Create launch template

Launch template name : **XYZ-WebServer-LaunchTemplate**

Template description : **Launch template for Auto Scaling Assignment**

AMI : My AMIs → Owned by me : **XYZ-WebServer-AMI , ami-027cde97e9d644dbc**

Instance type : **t3.micro**

Selected a keypair and left rest as default.

EC2 > Launch templates > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

XYZ-WebServer-LaunchTemplate

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

Launch template for Auto Scaling Assignment

Max 255 chars

Auto Scaling guidance

Info

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

▶ Template tags

▶ Source template

▼ Summary

Software Image (AMI)

AMI with Apache web server for...[read more](#)

ami-027cde97e9d644dbc

Virtual server type (instance type)

t3.micro

Firewall (security group)

-

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Create launch template

▼ Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Q Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

☐ Don't include in launch template

☒ Owned by me

☐ Shared with me

Q

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

XYZ-WebServer-AMI
ami-027cde97e9d644dbc
2026-09-28T04:52:51.000Z Virtualization: hvm ENA enabled: true
Root device type: ebs Boot mode: uefi-preferred

Description

AMI with Apache web server for Auto Scaling Assignment

Architecture	AMI ID
x86_64	ami-027cde97e9d644dbc

▼ Instance type [Info](#) | [Get advice](#) Advanced

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Linux base pricing: 0.0104 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour

On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand RHEL base pricing: 0.0392 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

mgia_keypair_1

▼

Create new key pair

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template

▼

Create new subnet [↗](#)

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone [Info](#)

Don't include in launch template

▼

Enable additional zones [↗](#)

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group

☐ Create security group

Security groups [Info](#)

Select security groups

▼

Compare security group rules

▶ Advanced network configuration

XYZ-WebServer-LaunchTemplate (lt-025c633158d58857d)

Actions ▼ Delete template

Launch template details

Launch template ID

lt-025c633158d58857d

Launch template name

XYZ-WebServer-LaunchTemplate

Default version

1

Owner

arn:aws:iam::430776404757:root

Details

Versions

Template tags

Launch template version details

Actions ▼ Delete template version

Version

1 (Default) ▼

Description

Launch template for Auto Scaling Assignment

Date created

2026-09-28T06:42:45.000Z

Created by

arn:aws:iam::430776404757:root

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-027cde97e9d644dbc

Instance type

t3.micro

Availability Zone

-

Availability Zone Id

-

Key pair name

mgia_keypair_1

Security groups

-

Security group IDs

-

To use this launch configuration to create an Auto Scaling group with 1 minimum and 3 maximum instances.

Task 2:

E. Created the Auto Scaling Group

AWS Console → EC2 → Auto Scaling Groups

Clicked Create Auto Scaling group

Group name: **XYZ-AutoScaling-Group**

Launch template: **XYZ-WebServer-LaunchTemplate**

Choose launch template Info

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
Enter a name to identify the group.

XYZ-AutoScaling-Group

Must be unique to this account in the current Region and no more than 255 characters.

Launch template Info

ⓘ For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

XYZ-WebServer-LaunchTemplate

[Create a launch template](#)

Version
Default (2)

[Create a launch template version](#)

Description XYZ-WebServer-LaunchTemplate	Launch template XYZ-WebServer-LaunchTemplate lt-025c633158d58857d	Instance type t3.micro
AMI ID ami-027cde97e9d644dbc	Security groups -	Request Spot Instances No
Key pair name mgia_keypair_1	Security group IDs sg-04191fc6a271cf3a4	

Additional details

Storage (volumes) -	Date created Mon Sep 28 2026 12:23:53 GMT+0530 (India Standard Time)
-------------------------------	---

Scaling

Info

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Max desired capacity

1

3

Equal or less than desired capacity

Equal or greater than desired capacity

Automatic scaling - optional

Choose whether to use a target tracking policy

Info

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

XYZ-AutoScaling-Group

XYZ-AutoScaling-Group Capacity overview

Edit

arn:aws:autoscaling:us-east-1:430776404757:autoScalingGroup:26b28c4d-98a5-4270-a7c2-a38353f57360:autoScalingGroupName/XYZ-AutoScaling-Group

Desired capacity

Scaling limits

Desired capacity type

Status

1

1 - 3

Units (number of instances)

Updating capacity

Date created

Mon Sep 28 2026 12:43:15 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template

lt-025c633158d58857d

XYZ-WebServer-LaunchTemplate

Version

Default

Description

XYZ-WebServer-LaunchTemplate

View details in the launch template console

AMI ID

ami-027cde97e9d644dbc

Security groups

-

Storage (volumes)

-

Instance type

t3.micro

Security group IDs

sg-04191fc6a271cf3a4

Key pair name

mgia_keypair_1

Owner

arn:aws:iam::430776404757:root

Create time

Mon Sep 28 2026 12:23:53 GMT+0530 (India Standard Time)

Request Spot Instances

No

Network

Edit

Availability Zones

use1-az2 (us-east-1b)

use1-az1 (us-east-1a)

Subnet ID

subnet-093a023aef10b9e26

subnet-02791cbf0e8268fb9

Availability Zone distribution

Balanced best effort

Successfully created autoscaling-group capacity 1-3 instances.

Verified the newly created instance

Auto Scaling Group name

XYZ-AutoScaling-Group

Instance summary for i-0bcea9ae31aab858b

Updated less than a minute ago

Connect

Instance state

Actions

Instance ID i-0bcea9ae31aab858b	Public IPv4 address 34.204.2.32 open address	Private IPv4 addresses 172.31.91.245
IPv6 address -	Instance state Running	Public DNS ec2-34-204-2-32.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-91-245.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-91-245.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 34.204.2.32 [Public IP]	VPC ID vpc-030682f2c586ad6b1	Auto Scaling Group name XYZ-AutoScaling-Group
IAM role -	Subnet ID subnet-093a023aef10b9e26	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:430776404757:instance/i-0bcea9ae31aab858b	
Operator -		

Used Public IP to check - webserver successfully launched.



XYZ-AutoScaling-Group

XYZ-AutoScaling-Group Capacity overview

Edit

arn:aws:autoscaling:us-east-1:430776404757:autoScalingGroup:26b28c4d-98a5-4270-a7c2-a38353f57360:autoScalingGroupName/XYZ-AutoScaling-Group

Desired capacity 1	Scaling limits 1 - 3	Desired capacity type Units (number of instances)	Status Updating capacity
-----------------------	-------------------------	--	-----------------------------

Date created
Mon Sep 28 2026 12:43:15 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

Launch template

Edit

Launch template lt-025c633158d58857d XYZ-WebServer-LaunchTemplate	AMI ID ami-027cde97e9d644dbc	Instance type t3.micro	Owner arn:aws:iam::430776404757:root
Version Default	Security groups -	Security group IDs sg-04191fc6a271cf3a4	Create time Mon Sep 28 2026 12:23:53 GMT+0530 (India Standard Time)
Description XYZ-WebServer-LaunchTemplate	Storage (volumes) -	Key pair name mgia_keypair_1	Request Spot Instances No

View details in the launch template console

Network

Edit

Availability Zones use1-az2 (us-east-1b) use1-az1 (us-east-1a)	Subnet ID subnet-093a023aef10b9e26 subnet-02791cbf0e8268fb9	Availability Zone distribution Balanced best effort
--	---	--

F. Created the Target Group

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.



Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB



IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB



Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB



Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

XYZ-ALB-TG

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 10/32

Protocol

Protocol for communication between the load balancer and targets.

HTTP

Port

Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type

Only targets with the indicated IP address type can be registered to this target group.



IPv4

Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.



IPv6

Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-030682f2c586ad6b1
172.31.0.0/16

(default)



[Create VPC](#)

Registered targets (1) [Info](#) Anomaly mitigation: Not applicable [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status detail
☐	i-03209d44401749ba2	-	80	us-east-1a (us...)	Unused	Target group is not c

Successfully created XYZ-ALB-TG

EC2 > Target groups > XYZ-ALB-TG

XYZ-ALB-TG

[Actions](#)

Details

arn:aws:elasticloadbalancing:us-east-1:430776404757:targetgroup/XYZ-ALB-TG/ae044357b3ea649b

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-030682f2c586ad6b1
IP address type IPv4	Load balancer None associated		

3 Total targets	0 Healthy	0 Unhealthy	3 Unused	0 Initial	0 Draining
	0 Anomalous				

G. Created and Tested Application Load Balancer

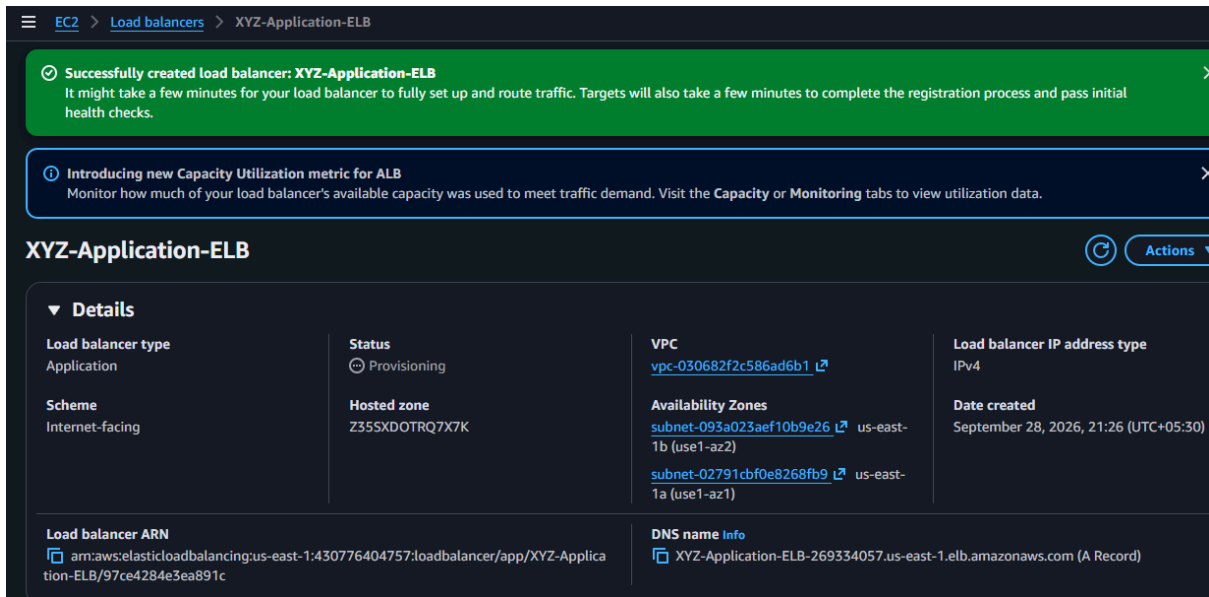
EC2 → Load Balancers → Create Load Balancer

Selected: Application Load Balancer. Clicked Create.

Configurations:

- Load Balancer Name: **XYZ-Application-ELB**
- Load Balancer Type: Application Load Balancer
- VPC: **vpc-030682f2c586ad6b1**
- Availability Zones/Subnets: Selected subnets in:
 - **us-east-1a**
 - **us-east-1b**
- Listener Protocol: HTTP, Listener Port: 80
- Forward traffic to **target group XYZ-ALB-TG**.

The Application Load Balancer successfully created.



Verified that the ALB became active.

Accessed the ALB DNS name using HTTP:

<http://xyz-application-elb-269334057.us-east-1.elb.amazonaws.com/>

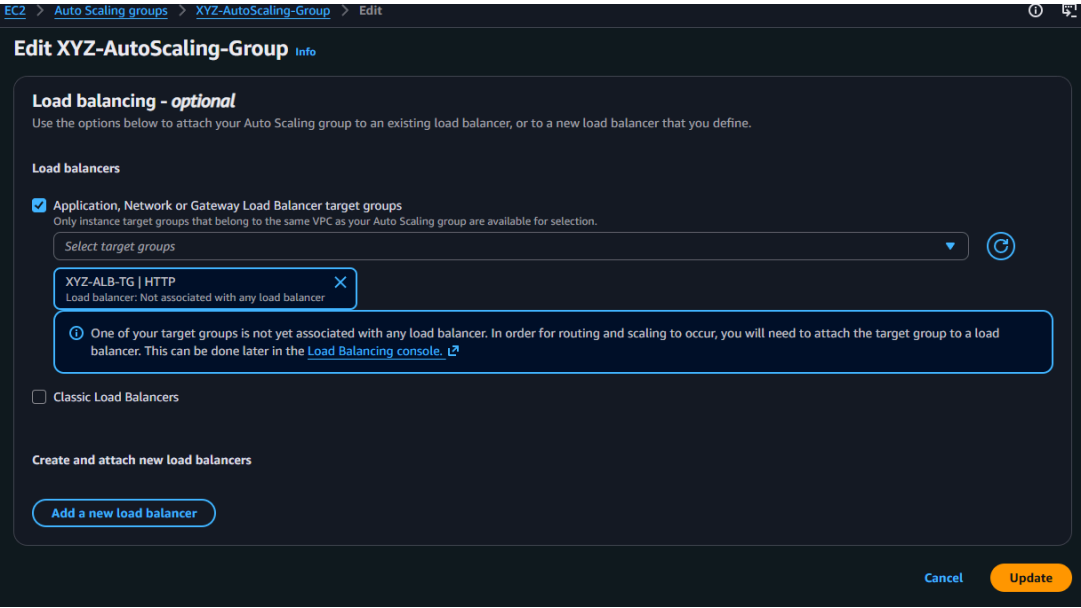
Verified that the Apache webpage was successfully displayed.



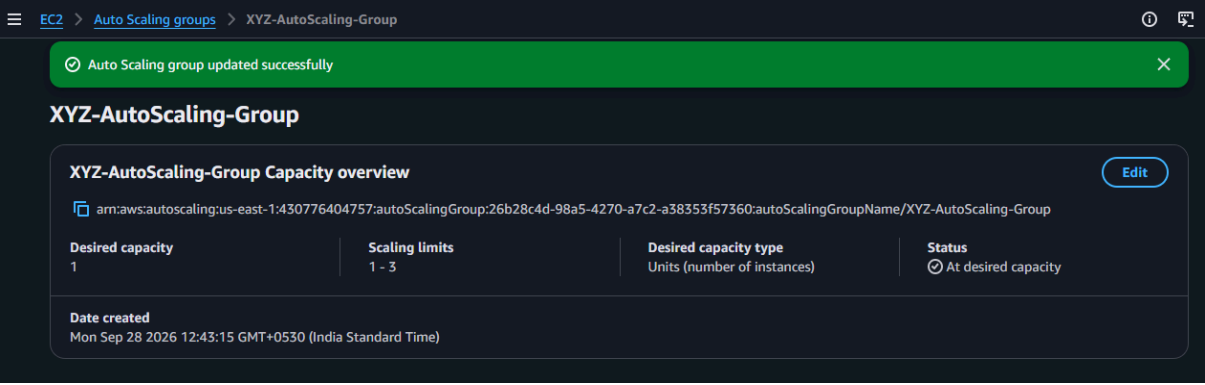
H. Attached the Target Group to the Auto Scaling Group

EC2 → Auto Scaling Groups → XYZ-AutoScaling-Group → Edit

Selected: XYZ-ALB-TG | HTTP



Clicked Update



Auto Scaling Group is successfully connected to Application Load Balancer-Target Group.

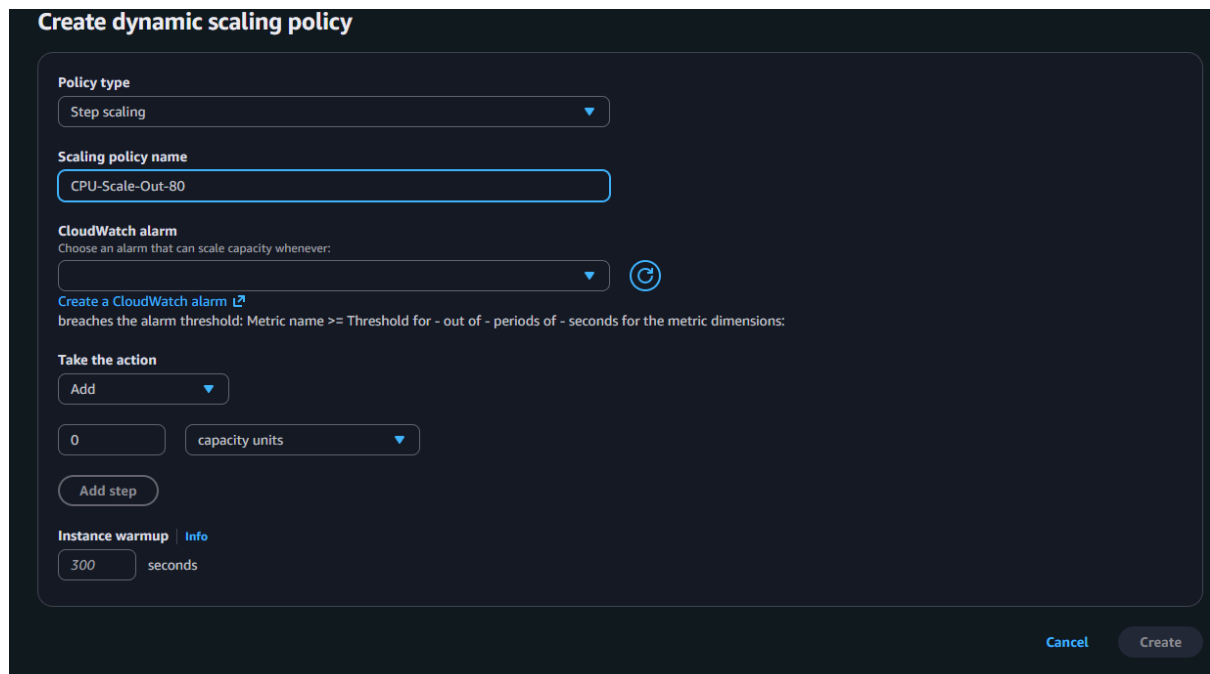
Task 3: Configured Auto Scale-In/Out

I. Scale out policy with CloudWatch CPU alarm :

Created a step scaling policy named CPU-Scale-Out-80 when CPU utilization exceeds 80% → add 1 EC2 instance.

steps:

- EC2 → Auto Scaling Groups → XYZ-AutoScaling-Group → Automatic scaling tab
- Clicked Create dynamic scaling policy
- Policy type: Step scaling
- Policy name: CPU-Scale-Out-80
- Clicked “Create a CloudWatch alarm”.



The screenshot shows the 'Create dynamic scaling policy' form in the AWS console. The form is titled 'Create dynamic scaling policy' and contains the following fields and options:

- Policy type:** A dropdown menu with 'Step scaling' selected.
- Scaling policy name:** A text input field containing 'CPU-Scale-Out-80'.
- CloudWatch alarm:** A dropdown menu with a 'Create a CloudWatch alarm' link and a refresh icon. Below the dropdown, it says 'breaches the alarm threshold: Metric name >= Threshold for - out of - periods of - seconds for the metric dimensions:'.
- Take the action:** A dropdown menu with 'Add' selected.
- 0** (input field) **capacity units** (dropdown menu).
- Add step** (button).
- Instance warmup:** A text input field containing '300' followed by 'seconds'.

At the bottom right, there are 'Cancel' and 'Create' buttons.

Created a CloudWatch alarm

CloudWatch alarm for CPUUtilization greater than 80% :

Name: **CPU-Alarm-Scale-Out-80**

- On the CloudWatch metric selection:
- **EC2 → By Auto Scaling Group**
- XYZ-AutoScaling-Group: CPUUtilization
- **Statistic:** Average
- **Period:** 1 minute
- **Threshold type:** Static
- **CPUUtilization:** Greater than 80%
- **Datapoints to alarm:** 1 of 1

Metric

Namespace

AWS/EC2

Metric name

CPUUtilization

AutoScalingGroupName

XYZ-AutoScaling-Group

Statistic

Average

Period

1 minute

Conditions

Threshold type

Static

Use a value as a threshold

Anomaly detection

Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

Greater

> threshold

Greater/Equal

>= threshold

Lower/Equal

<= threshold

Lower

< threshold

than...

Define the threshold value.

80

Must be a number.

Additional configuration

Preview

1h 3h 12h 1d 3d 1w Custom

UTC timezone

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

Percent

80%

80

40.17%

0.33%

13:30 14:00 14:30 15:00 15:30 16:00

CPUUtilization

Selected the created alarm and **the scale-out policy successfully created.**

Create dynamic scaling policy

Policy type

Step scaling

Scaling policy name

CPU-Scale-Out-80

CloudWatch alarm

Choose an alarm that can scale capacity whenever:

CPU-Alarm-Scale-Out-80

Create a CloudWatch alarm

breaches the alarm threshold: CPUUtilization > 80 for 1 out of 1 periods of 60 seconds for the metric dimensions:

AutoScalingGroupName = XYZ-AutoScaling-Group

Take the action

Add

0

capacity units

when

80

<=

CPUUtilization

<

+infinity

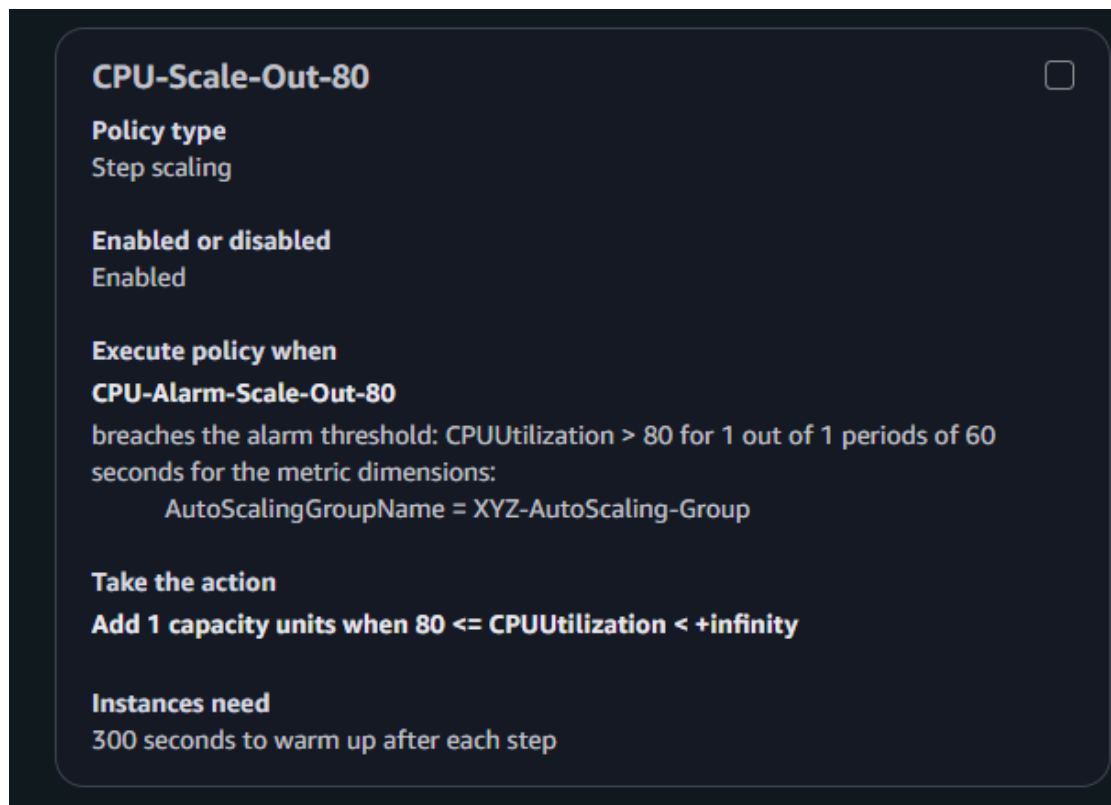
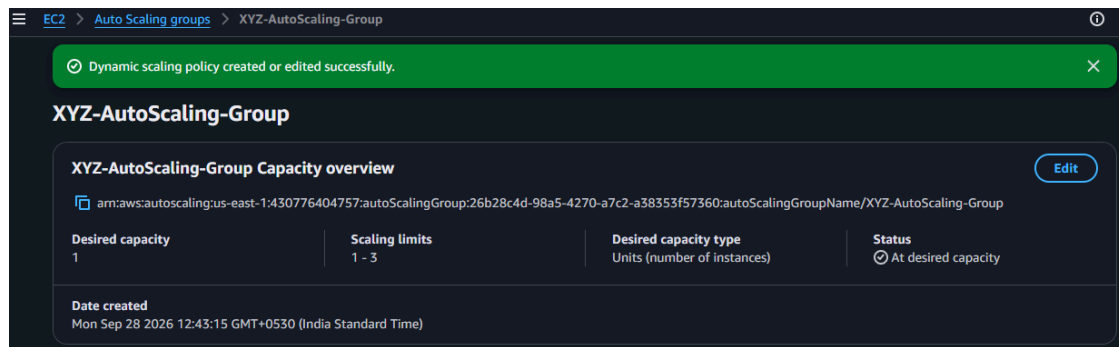
Add step

Instance warmup

Info

300

seconds



J. Scale in policy with CloudWatch CPU alarm :

Created a step scaling policy named CPU-Scale-In-60 when CPU utilization is below 60% → remove 1 EC2 instance.

steps:

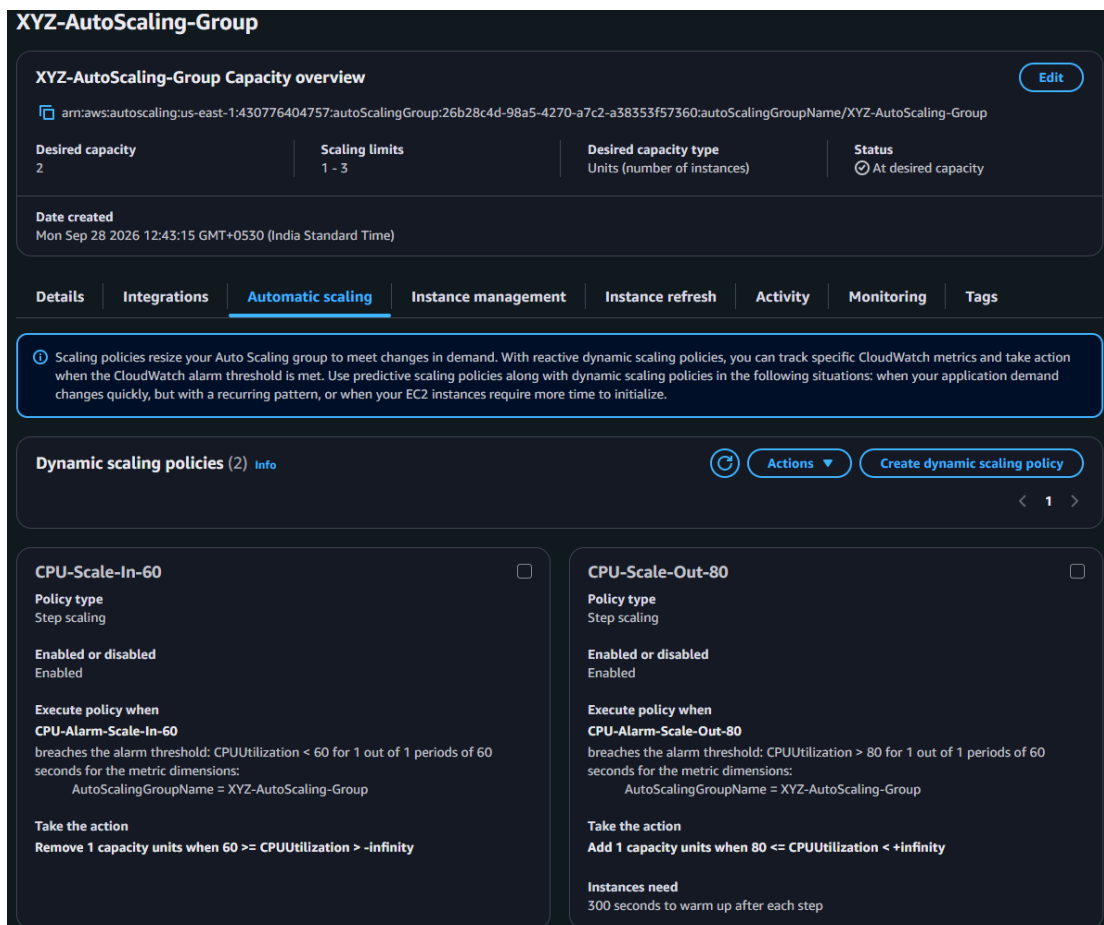
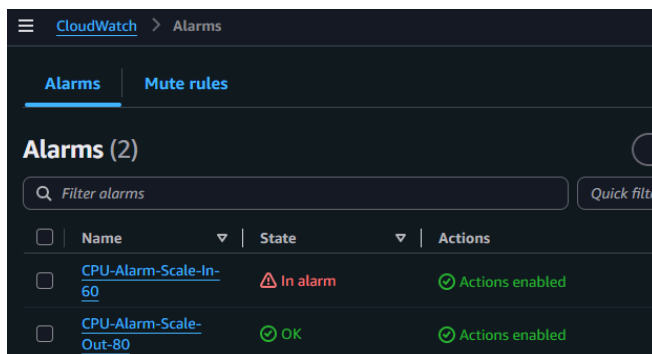
- EC2 → Auto Scaling Groups → XYZ-AutoScaling-Group → Automatic scaling tab
- Clicked Create dynamic scaling policy
- Policy type: Step scaling
- Policy name: CPU-Scale-In-60
- Clicked “Create a CloudWatch alarm”.

Created a CloudWatch alarm

CloudWatch alarm for CPUUtilization less than 60% :

Name: **CPU-Alarm-Scale-In-60**

- On the CloudWatch metric selection:
- **EC2 → By Auto Scaling Group**
- XYZ-AutoScaling-Group: CPUUtilization
- **Statistic:** Average
- **Period:** 1 minute
- **Threshold type:** Static
- **CPUUtilization: less than 60%**
- **Datapoints to alarm:** 1 of 1



Both scaling policies are successfully created.

Task 4: Routing to Domain

Route 53 → Application Load Balancer

K. Created a public Route 53 hosted zone.

AWS Console → Route 53 → Hosted zones → Create hosted zone

Configurations:

Domain name: **xyzawsweb.duckdns.org**

Type: Public hosted zone

Successfully created hosted zone.

L. Added A record to Route 53 hosted zone

Configurations:

Record type: A

Alias: ON

Route traffic to: Application Load Balancer

Region: US East (N. Virginia)

Load balancer: XYZ-Application-ELB

Routing policy: Simple

The screenshot shows the 'Create record' page in the AWS Route 53 console. The page has a dark theme. At the top, it says 'Create record' with an 'Info' link. Below this is a 'Quick create record' section. On the right of this section is a 'Switch to wizard' link. Under 'Record 1', there is a 'Delete' button. The 'Record name' field is set to 'subdomain' and the 'Record type' is set to 'A - Routes traffic to an IPv4 address and some AWS resources'. Below the record name, there is a note: 'Keep blank to create a record for the root domain.' The 'Alias' checkbox is checked. The 'Route traffic to' section shows 'Alias to Application and Classic Load Balancer', 'US East (N. Virginia)', and the target 'dualstack.XYZ-Application-ELB-269334057.us-east-1.elb.amazonaws.com'. Below this, the 'Alias hosted zone ID' is 'Z35SXDOTRQ7X7K'. The 'Routing policy' is set to 'Simple routing' and 'Evaluate target health' is checked. At the bottom right, there are 'Add another record', 'Cancel', and 'Create records' buttons.

Create record [Info](#)

Quick create record [Switch to wizard](#)

▼ Record 1 [Delete](#)

Record name [Info](#) xyzawsweb.duckdns.org Record type [Info](#) A - Routes traffic to an IPv4 address and some AWS resources

subdomain

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to [Info](#)

Alias to Application and Classic Load Balancer

US East (N. Virginia)

Q dualstack.XYZ-Application-ELB-269334057.us-east-1.elb.amazonaws.com X

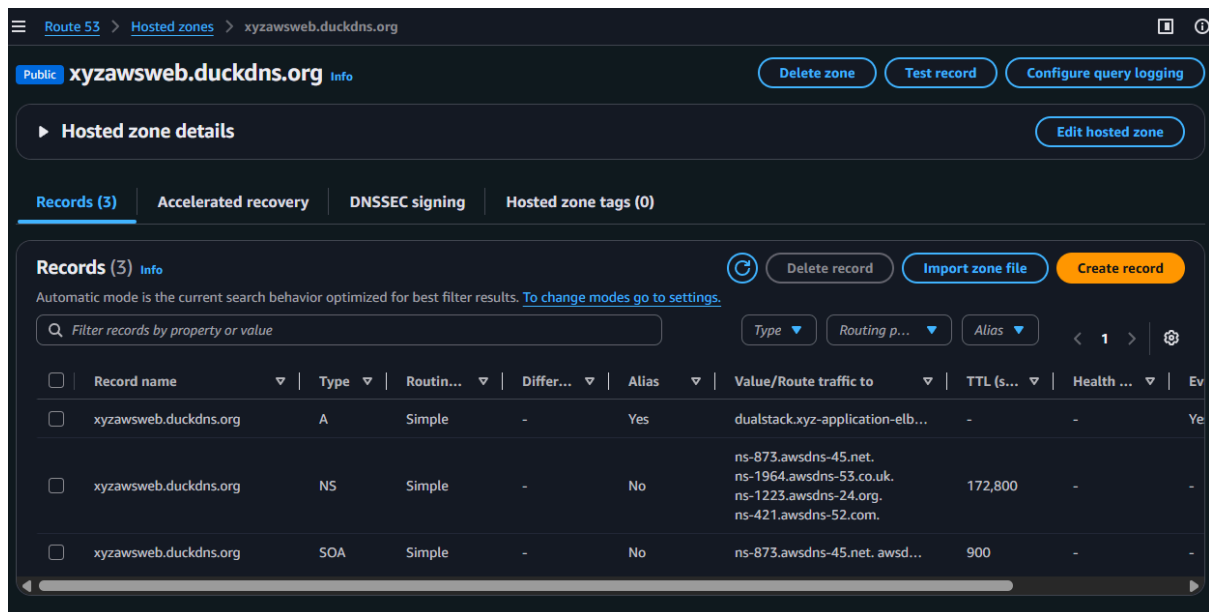
Alias hosted zone ID: Z35SXDOTRQ7X7K

Routing policy [Info](#) Evaluate target health

Simple routing ☒ Yes

[Add another record](#)

[Cancel](#) [Create records](#)

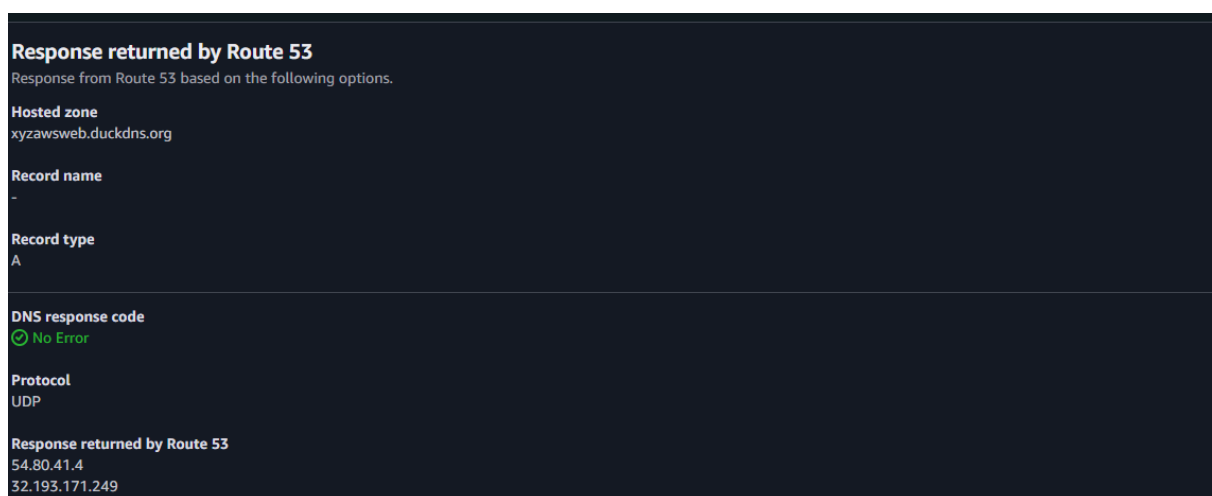


Task 5: Verification

Verified the Route 53 record

- Route 53 → Hosted zones → Opened the hosted zone.
- Selected the **A Alias** record.
- Clicked **Test record**.
- Record type:** A → Clicked **Get response**.

The result showed a successful DNS response with response code: No Error.
Route 53 returned **two IPv4 addresses: 54.80.41.4 and 32.193.171.249**



The ALB alias may return **multiple IPv4 addresses**, which is normal for an ALB; your assignment evidence already records two returned addresses.

Verified the instance

Instance summary for i-0aabf7833e5d321b0

Updated less than a minute ago

Instance ID

i-0aabf7833e5d321b0

IPv6 address

-

Hostname type

IP name: ip-172-31-91-40.ec2.internal

Answer private resource DNS name

-

Auto-assigned IP address

98.94.43.246 [Public IP]

IAM role

-

IMDSv2

Required

Operator

-

Public IPv4 address

98.94.43.246 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-91-40.ec2.internal

Instance type

t3.micro

VPC ID

vpc-030682f2c586ad6b1

Subnet ID

subnet-093a023aef10b9e26

Instance ARN

arn:aws:ec2:us-east-1:430776404757:instance/i-0aabf7833e5d321b0

Private IPv4 addresses

172.31.91.40

Public DNS

ec2-98-94-43-246.compute-1.amazonaws.com | open address

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

XYZ-AutoScaling-Group

Managed

false

The instance confirms the EC2 instance is currently Running and is associated with: XYZ-AutoScaling-Group.

Verified the Application Load Balancer

EC2 → Load Balancers → XYZ-Application-ELB

Confirmed:

Status: Active

Listener: HTTP : 80

Default action: Forward → XYZ-ALB-TG

XYZ-Application-ELB

Actions

Details

Load balancer type

Application

Scheme

Internet-facing

Status

Active

Hosted zone

Z35SXDOTRQ7X7K

VPC

vpc-030682f2c586ad6b1

Availability Zones

subnet-093a023aef10b9e26 us-east-1b (use1-az2)
subnet-02791cbf0e8268fb9 us-east-1a (use1-az1)

Load balancer IP address type

IPv4

Date created

September 28, 2026, 21:26 (UTC+05:30)

Load balancer ARN

arn:aws:elasticloadbalancing:us-east-1:430776404757:loadbalancer/app/XYZ-Application-ELB/97ce4284e3ea891c

DNS name

XYZ-Application-ELB-269334057.us-east-1.elb.amazonaws.com (A Record)

Listeners and rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

Listeners and rules (1)

Manage rules

Manage listener

Add listener

A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.

Filter listeners

< 1 >

Protocol:Port

Default action

Rules

ARN

Security policy

Default SSL/TLS certificate

HTTP:80

Forward to target group
XYZ-ALB-TG 1 (100%)
Target group stickiness: Off

1 rule

ARN

Not applicable

Not applicable

Verified the Target Group

Confirmed the current ASG instance shows: Healthy

XYZ-ALB-TG

Details

arn:aws:elasticloadbalancing:us-east-1:430776404757:targetgroup/XYZ-ALB-TG/ae044357b3ea649b

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-030682f2c586ad6b1
IP address type IPv4	Load balancer XYZ-Application-ELB		

1 Total targets	1 Healthy 0 Anomalous	0 Unhealthy	0 Unused	0 Initial	0 Draining
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Verified Auto Scaling

Automatic scaling should shows two dynamic scaling policies:

CPU > 80% → Add 1

CPU < 60% → Remove 1

Scaling Minimum = 1, Desired = 1, Maximum = 3

XYZ-AutoScaling-Group

XYZ-AutoScaling-Group Capacity overview

arn:aws:autoscaling:us-east-1:430776404757:autoScalingGroup:26b28c4d-98a5-4270-a7c2-a38353f57360:autoScalingGroupName/XYZ-AutoScaling-Group

Desired capacity 1	Scaling limits 1 - 3	Desired capacity type Units (number of instances)	Status At desired capacity
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Date created
Mon Sep 28 2026 12:43:15 GMT+0530 (India Standard Time)

Details

Integrations

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Tags

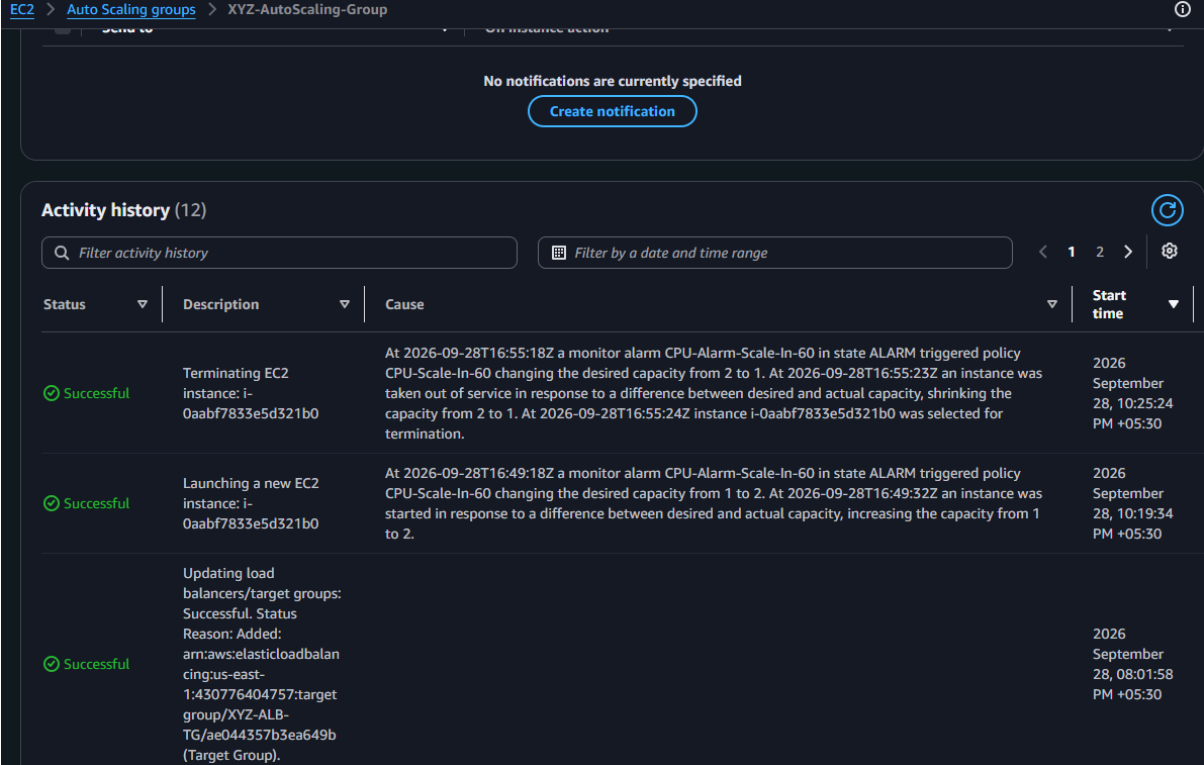
Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, you can track specific CloudWatch metrics and take action when the CloudWatch alarm threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when your application demand changes quickly, but with a recurring pattern, or when your EC2 instances require more time to initialize.

Dynamic scaling policies (2)

CPU-Scale-In-60
Policy type
Step scaling
Enabled or disabled
Enabled
Execute policy when
CPU-Alarm-Scale-In-60
breaches the alarm threshold: CPUUtilization < 60 for 1 out of 1 periods of 60 seconds for the metric dimensions:
AutoScalingGroupName = XYZ-AutoScaling-Group
Take the action
Remove 1 capacity units when 60 >= CPUUtilization > -infinity

CPU-Scale-Out-80
Policy type
Step scaling
Enabled or disabled
Enabled
Execute policy when
CPU-Alarm-Scale-Out-80
breaches the alarm threshold: CPUUtilization > 80 for 1 out of 1 periods of 60 seconds for the metric dimensions:
AutoScalingGroupName = XYZ-AutoScaling-Group
Take the action
Add 1 capacity units when 80 <= CPUUtilization < +infinity
Instances need
300 seconds to warm up after each step

XYZ-AutoScaling-Group is successfully scaling in/out instances.



The screenshot shows the AWS Management Console for the 'XYZ-AutoScaling-Group'. At the top, it states 'No notifications are currently specified' with a 'Create notification' button. Below is the 'Activity history (12)' section, which includes a search bar and a table of activities. The table has columns for Status, Description, Cause, and Start time. Three activities are visible, all with a 'Successful' status.

Status	Description	Cause	Start time
Successful	Terminating EC2 instance: i-0aabf7833e5d321b0	At 2026-09-28T16:55:18Z a monitor alarm CPU-Alarm-Scale-In-60 in state ALARM triggered policy CPU-Scale-In-60 changing the desired capacity from 2 to 1. At 2026-09-28T16:55:23Z an instance was taken out of service in response to a difference between desired and actual capacity, shrinking the capacity from 2 to 1. At 2026-09-28T16:55:24Z instance i-0aabf7833e5d321b0 was selected for termination.	2026 September 28, 10:25:24 PM +05:30
Successful	Launching a new EC2 instance: i-0aabf7833e5d321b0	At 2026-09-28T16:49:18Z a monitor alarm CPU-Alarm-Scale-In-60 in state ALARM triggered policy CPU-Scale-In-60 changing the desired capacity from 1 to 2. At 2026-09-28T16:49:32Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 1 to 2.	2026 September 28, 10:19:34 PM +05:30
Successful	Updating load balancers/target groups: Successful. Status Reason: Added: arn:aws:elasticloadbalancing:us-east-1:430776404757:target-group/XYZ-ALB-TG/ae044357b3ea649b (Target Group).		2026 September 28, 08:01:58 PM +05:30

Public DNS lookup verification

nslookup was used to verify the public DNS response for xyzawsweb.duckdns.org. The result was a non-authoritative DNS response. Route 53's own Test Record confirmed that the configured A Alias record returns the Application Load Balancer addresses.

```
Administrator: Windows PowerShell

Windows PowerShell
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PS C:\Windows\system32> nslookup xyzawsweb.duckdns.org
Server:      UnKnown
Address:     fe80::2ec1:f4ff:fe88:6241

Non-authoritative answer:
Name:       xyzawsweb.duckdns.org
Address:    3.223.127.201

PS C:\Windows\system32>
```